

EXPERIMENT :19

Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

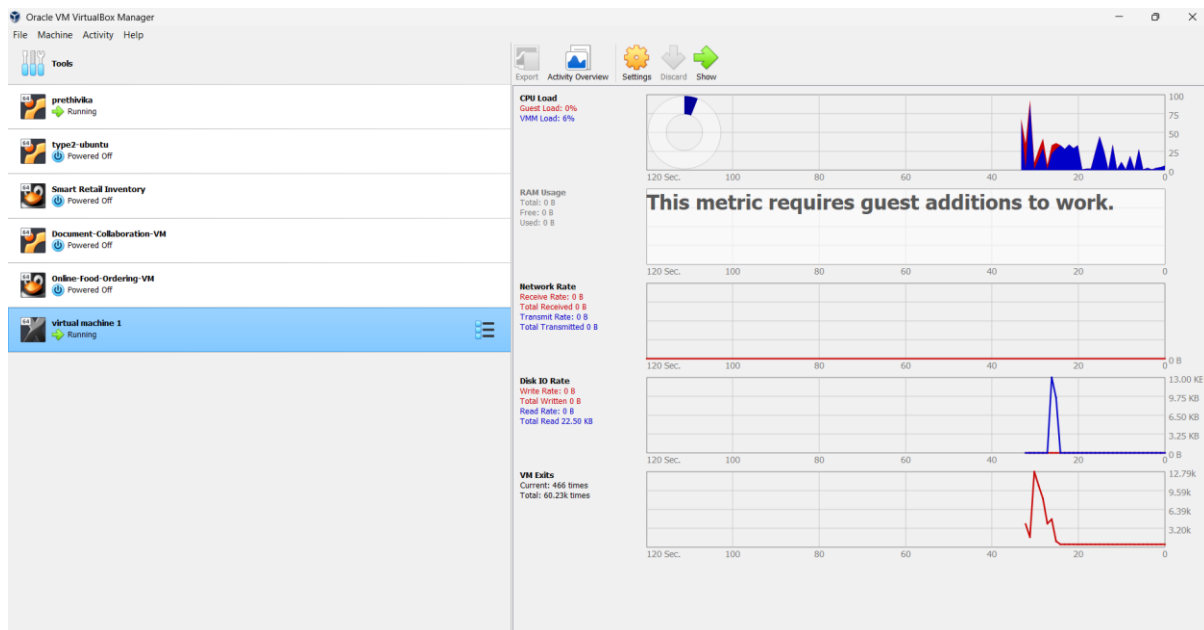
AIM:

To install the GCC C compiler in an Ubuntu virtual machine created using Oracle VirtualBox and compile and execute a simple C program to determine whether a given number is even or odd.

PROCEDURE:

1. Open Oracle VirtualBox, start the existing Ubuntu Virtual Machine, and log in to the Ubuntu desktop.
2. Open the Terminal and update the package repository using `sudo apt update`.
3. Install the GCC C compiler and required development tools using `sudo apt install build-essential`.
4. Verify the compiler installation using `gcc --version` and confirm that GCC is available in the Ubuntu virtual machine.
5. Create a C source file using a text editor such as Nano and enter the Even and Odd Number program using the modulus (%) operator.
6. Compile the program using `gcc filename.c -o filename` and execute it using `./filename`, then enter a number when prompted.
7. Verify the displayed output and confirm whether the entered number is Even or Odd, demonstrating successful C program compilation and execution inside the Ubuntu virtual machine.

IMPLEMENTATION:



```
Activities Terminal Thu 16:05 en1
ubuntu@ubuntu: ~/c-programs
File Edit View Search Terminal Help
ubuntu@ubuntu:~$ gcc --version
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 18.04.3 LTS _Bionic Beaver_ - Release amd64 (20190805) bio
nic InRelease
Hit:2 cdrom://Ubuntu 18.04.3 LTS _Bionic Beaver_ - Release amd64 (20190805) bio
nic Release
Hit:4 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:5 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:7 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [3,09
2 kB]
Get:8 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packages [2,
763 kB]
Get:9 http://archive.ubuntu.com/ubuntu bionic-updates/main Translation-en [552
kB]
Get:10 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 DEP-11 Metada
ta [297 kB]
Get:11 http://archive.ubuntu.com/ubuntu bionic-updates/main DEP-11 48x48 Icons
[82.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates/main DEP-11 64x64 Icons
[154 kB]
Get:13 http://archive.ubuntu.com/ubuntu bionic-updates/restricted amd64 Package
s [1,368 kB]
Get:14 http://security.ubuntu.com/ubuntu bionic-security/main Translation-en [4
```

```
Activities Terminal Thu 16:05 en1
ubuntu@ubuntu: ~/c-programs
File Edit View Search Terminal Help
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

ubuntu@ubuntu:~$ which gcc
/usr/bin/gcc
ubuntu@ubuntu:~$ mkdir c-programs
ubuntu@ubuntu:~$ cd c-programs
ubuntu@ubuntu:~/c-programs$ pwd
/home/ubuntu/c-programs
ubuntu@ubuntu:~/c-programs$ nano hello.c
ubuntu@ubuntu:~/c-programs$ ls
hello.c
ubuntu@ubuntu:~/c-programs$ cat hello.c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    if(n%2==0){
        printf("even num");
    }
    else{
        printf("odd num");
    }
    return 0;
}
ubuntu@ubuntu:~/c-programs$ gcc hello.c -o hello
ubuntu@ubuntu:~/c-programs$ ls
```

```
Activities Terminal Thu 16:06 en1
ubuntu@ubuntu: ~/c-programs
File Edit View Search Terminal Help
ubuntu@ubuntu:~$ mkdir c-programs
ubuntu@ubuntu:~$ cd c-programs
ubuntu@ubuntu:~/c-programs$ pwd
/home/ubuntu/c-programs
ubuntu@ubuntu:~/c-programs$ nano hello.c
ubuntu@ubuntu:~/c-programs$ ls
hello.c
ubuntu@ubuntu:~/c-programs$ cat hello.c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    if(n%2==0){
        printf("even num");
    }
    else{
        printf("odd num");
    }
    return 0;
}
ubuntu@ubuntu:~/c-programs$ gcc hello.c -o hello
ubuntu@ubuntu:~/c-programs$ ls
hello hello.c
ubuntu@ubuntu:~/c-programs$ ./hello
10
even numubuntu@ubuntu:~/c-programs$ ./hello
5
odd numubuntu@ubuntu:~/c-programs$
```

RESULT:

The GCC C compiler was successfully installed in the Ubuntu virtual machine using Oracle VirtualBox. The Even and Odd Number C program was successfully compiled and executed, and the program correctly identified the given number as either **even or odd**.